WORKSHEET 8.4

Area Between Curves, Slicing Vertically

Unit 8: Applications of Integration

Key ideas

- The area between two graphs for $a \le x \le b$ is $\displaystyle\int_a^b (\text{top} - \text{bottom})\,dx$.
- The limits are usually the x-coordinates of the points of intersection: set the two formulas equal.
- Top minus bottom is never negative, so an area found this way is positive even when the region lies below the x-axis.
- Sketch first. A test value between the intersections shows which graph is on top.

Worked examples

Example 1 · A line and a parabola

Find the area enclosed by $y = x^2 - 4x$ and $y = 2x - 5$.

Solution. $x^2 - 6x + 5 = 0$ gives $x = 1, 5$. The line is on top: $\displaystyle\int_1^5 (-x^2 + 6x - 5)\,dx$.

$\dfrac{32}{3}$.

Example 2 · Given limits

Find the area between $y = e^x$ and $y = x$ for $0 \le x \le 2$.

Solution. $\displaystyle\int_0^2 (e^x - x)\,dx = \left[e^x - \tfrac{x^2}{2}\right]_0^2$.

$e^2 - 3$.

Example 3 · Finding the limits

Find the area enclosed by $y = 3\sqrt{x}$ and $y = x$.

Solution. They meet at $x = 0$ and $x = 9$: $\displaystyle\int_0^9 \left(3\sqrt{x} - x\right)\,dx = 54 - \frac{81}{2}$.

$\dfrac{27}{2}$.

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Practice

CORE

1. Write, but do not evaluate, an integral for the area enclosed by $y = x + 2$ and $y = x^2$.

2. Evaluate the integral in Question 1.

3. Find the area enclosed by $y = x^2$ and $y = 4$.

4. Find the area between $y = x$ and $y = x^3$ for $0 \leq x \leq 1$.

EXAM LEVEL

5. Find the area enclosed by $y = x^2$ and $y = 8 - x^2$.

6. Find the area between $y = \cos x$ and $y = \sin x$ for $0 \leq x \leq \dfrac{\pi}{4}$.

7. Find the area between $y = e^x$ and $y = e^{-x}$ for $0 \leq x \leq 1$.

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8. *Calculator.* Find the area of the region in the first quadrant enclosed by $y = 3\sin x$ and $y = x$.

Question 9 · Free-response practice

Let $f(x) = 5x - x^2$ and $g(x) = x$. Let R be the region enclosed by the graphs of f and g.

(a) Find the x-coordinates of the points of intersection of the two graphs.

(b) Find the area of R.

(c) Find the value of x at which the vertical distance between the two graphs within R is greatest, and find that distance.

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Worked solutions

1. $\displaystyle\int_{-1}^{2}(x+2-x^2)\,dx$.

2. $\dfrac{9}{2}$.

3. $\displaystyle\int_{-2}^{2}(4-x^2)\,dx = \frac{32}{3}$.

4. $\dfrac{1}{2}-\dfrac{1}{4}=\dfrac{1}{4}$.

5. They meet at $x=\pm 2$: $\displaystyle\int_{-2}^{2}(8-2x^2)\,dx = \frac{64}{3}$.

6. $\Big[\sin x+\cos x\Big]_0^{\pi/4} = \sqrt{2}-1$.

7. $\Big[e^x+e^{-x}\Big]_0^1 = e+\dfrac{1}{e}-2$.

8. They meet at $x=0$ and $x\approx 2.279$: $\displaystyle\int_0^{2.279}(3\sin x - x)\,dx \approx 2.354$.

9. (a) $5x-x^2=x$ gives $x=0$ and $x=4$.

   (b) $\displaystyle\int_0^4(4x-x^2)\,dx = 32-\frac{64}{3}=\frac{32}{3}$.

   (c) The distance is $4x-x^2$, with derivative $4-2x$ changing from positive to negative at $x=2$. The greatest distance is 4.

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